

Final Capstone Project
Advanced Business Analytics
BIT-3484 (T/TH 11:00)
May 5th, 2026

Introduction

In today's platform-based economy, user-generated ratings play a critical role in shaping business success, particularly in marketplaces like Airbnb. On Airbnb, guest ratings directly influence a listing's visibility, booking volume, and pricing power, making them one of the most important performance metrics for hosts. Higher-rated listings are more likely to appear in search results, attract consistent demand, and command premium prices. As a result, understanding what drives guest satisfaction is essential for hosts seeking to improve their performance, as well as for the platform itself, which relies on high-quality experiences to maintain trust and long-term growth.

In the short-term rental industry, competition is high and guests have a wide range of options across locations, price points, and property types. Because of this, guests can easily choose a different listing if their expectations are not met. This makes ratings especially important, as even small differences in perceived quality can significantly impact booking decisions. Prior research has shown that online reviews and ratings play a significant role in shaping consumer decisions on Airbnb, where trust between hosts and guests is a key factor in booking behavior (Zervas et al. 2017). However, guest satisfaction is influenced by a combination of factors, including listing characteristics, host behavior, pricing, and location. Additionally, research suggests that Airbnb ratings tend to be consistently high, indicating that even small differences in ratings can meaningfully impact how listings are perceived by potential guests (Zervas et al., 2015). Without a clear understanding of which of these factors matter most, hosts may struggle to prioritize improvements effectively and may invest time and resources into changes that do not meaningfully impact ratings.

This analysis addresses that challenge by using Airbnb listing data from Inside Airbnb to identify the key drivers of guest ratings. Specifically, it seeks to answer three central research questions: what listing and host characteristics are most strongly associated with higher ratings; whether more expensive listings receive higher ratings or are evaluated more critically due to elevated guest expectations; and which factors influencing ratings are within a host's control, such as cleanliness and responsiveness, versus fixed factors like location and neighborhood.

Beyond the technical analysis, we are particularly interested in how data can shape public perception and influence decision-making. On platforms like Airbnb, even small differences in ratings or reviews can significantly affect how a listing is viewed, despite minimal differences in quality. This highlights the power of data not just as a measurement tool, but as a driver of perception and behavior. We are passionate about using data to uncover insights that can help both businesses and individuals make more informed decisions. Specifically, this project allows us to explore how data-driven analysis can guide hosts toward actionable improvements in their listings, helping them enhance guest experiences,

increase bookings, and grow their business. This interest in using data to create practical, real-world impact is a key reason we chose to explore this topic.

Data Source (EDA & Preprocessing)

The dataset was sourced from [Inside Airbnb](#), an independent open-source project that scrapes and publishes publicly available Airbnb listing data. The specific extract used covers one city and consists of seven files: a detailed listings file (listings.csv.gz), a summary listings file (listings.csv), a calendar file (calendar.csv.gz), two reviews files (reviews.csv and reviews.csv.gz), and neighbourhood boundary files (neighbourhoods.csv and neighbourhoods.geojson). All analysis was conducted on the detailed listings file, which contains 85 columns per listing covering host attributes, property characteristics, availability, and guest review scores.

The raw dataset contained 453 listings, of which 395 (87.2%) had at least one guest review and a valid overall rating. Only rated listings were used in the EDA and modeling phases, as unrated listings provide no signal for the target variable. Several preprocessing steps were applied before analysis. The amenity count was derived from the raw amenities field, which stores amenities as a JSON-formatted string array. Rather than parsing the JSON directly, the number of amenities was computed by counting quoted string tokens within the field using a regular expression, producing a single numeric feature representing listing amenity richness. Boolean fields such as superhost status (host_is_superhost), host identity verification (host_identity_verified), and instant bookability (instant_bookable) were stored as "t"/"f" string literals in the raw data and were remapped to binary integer values (1/0) for use in correlation analysis and modeling. Host tenure was available as a pre-computed numeric field (hosts_time_as_host_years) and was used directly.

Three columns that would have been analytically valuable are price, host_response_time, and host_response_rate; however, they were entirely null in this extract, likely suppressed by Inside Airbnb for this city's release. An audit was conducted and confirmed that price was null across all 453 listings in both the summary and detailed listings files, and across all 165,345 rows of the calendar file including the adjusted_price field. This ruled out any direct price-based analysis and necessitated a revised approach to Research Question 2, discussed below.

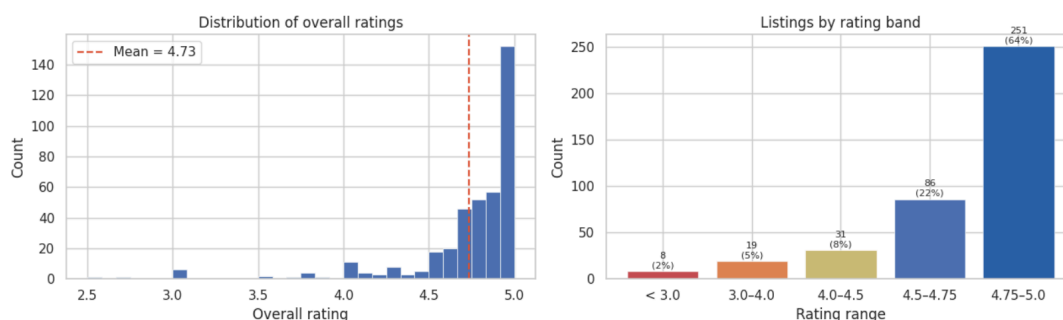


Figure 1. Distribution of overall ratings across 395 rated listings (left) and breakdown by rating band (right). The strong left skew and concentration in the 4.75–5.0 band reflect well-documented rating inflation on the Airbnb platform, and motivated the use of ≥ 4.75 as the high-rated threshold in modeling.

Overall ratings were heavily left-skewed, with a mean of 4.73 out of 5.0. As shown in Figure 1, 63.5% of listings scored between 4.75 and 5.0, and only 6.8% scored below 4.0. This compression in the rating scale means meaningful quality differentiation occurs almost entirely within a narrow half-point band, which has direct implications for both EDA interpretation and the choice of binary classification threshold in the modeling phase.

Research Questions

RQ 1. What listing and host characteristics are most associated with higher ratings?

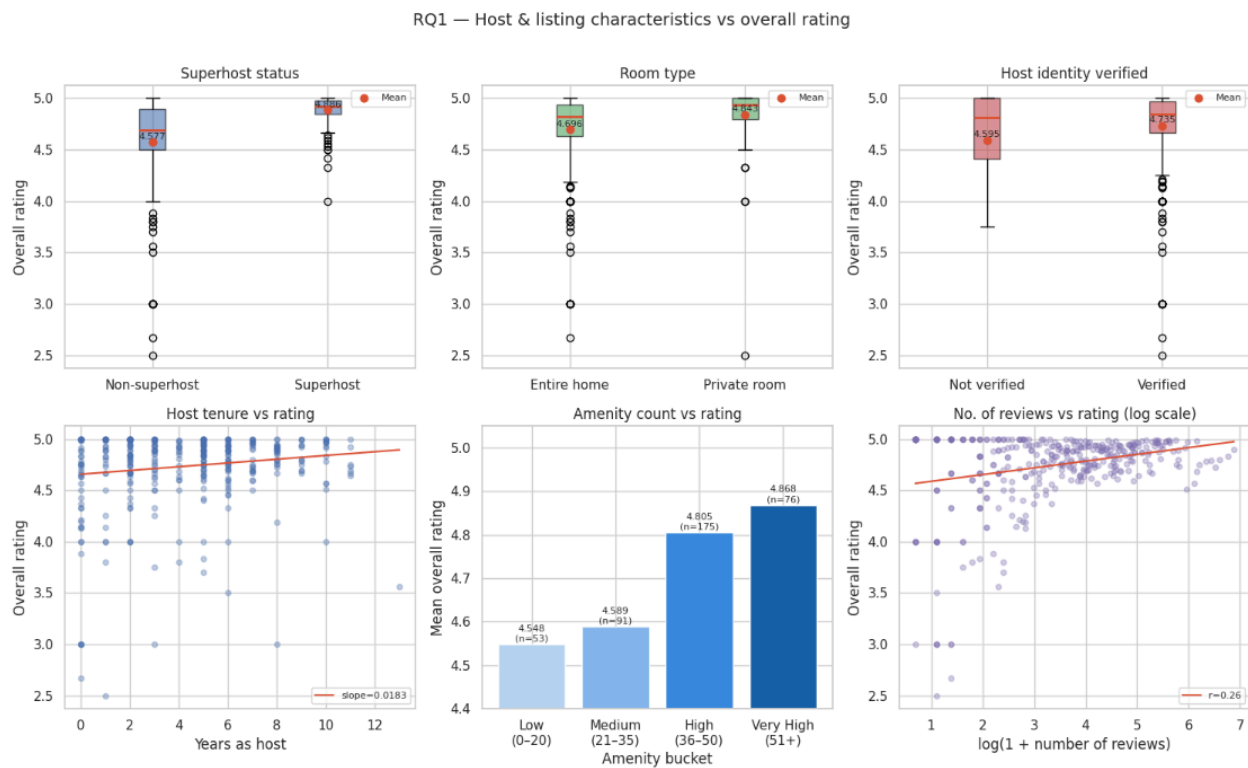
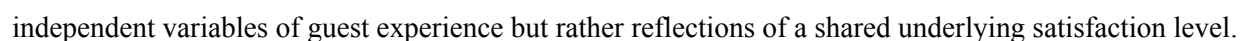


Figure 2. Host and listing characteristics vs overall rating. Top row: categorical comparisons (superhost status, room type, host identity verification). Bottom row: numeric relationships (host tenure, amenity count by bucket, number of reviews on log scale).

Superhost status was the single largest categorical predictor of overall rating. Superhosts averaged 4.886 compared to 4.577 for non-superhosts. That is a 0.31-point gap that is substantial given Airbnb's compressed rating scale, and one that persisted across every sub-rating dimension. This likely reflects the compounding effect of superhost requirements: hosts must maintain high response rates, low cancellation rates, and consistent cleanliness standards to qualify, meaning superhost status functions as a proxy for overall hosting quality rather than any single behaviour.

Host tenure had a near-zero slope against overall rating, indicating that years of experience alone do not drive guest satisfaction. A host's longevity on the platform does not appear to have any measurable rating advantage, suggesting that quality is determined more by current hosting behaviours than accumulated experience. Review count showed a modest positive correlation ($r = 0.15$ on a log scale), consistent with a selection effect: higher-rated listings attract more bookings and therefore accumulate

The correlation heatmap in Figure 3 confirms these patterns at a glance and reveals an additional finding: the six sub-ratings are all strongly intercorrelated with one another, suggesting they are not



Value ($r = 0.86$) and accuracy ($r = 0.84$) were the strongest individual predictors of overall rating, meaning guests who felt a listing did not deliver on its implied price point or its listing description were the most likely to leave a lower overall score. Communication ($r = 0.75$) and cleanliness ($r = 0.82$) were also strong, while location ($r = 0.68$) was the weakest sub-rating predictor, consistent with the idea that guests mentally separate the fixed physical location from their assessment of the host's performance.

RQ 2. Do listings that indicate they would be more expensive get rated higher (room type, room size, amenities), or do guests have higher expectations and rate more critically?

The dataset, unfortunately, does not directly show us the pricing for each Airbnb unit. To address this, we analyzed features that typically indicate a more expensive rental: room type, room size, and number of amenities. First, we analyzed the relationship between room type and guest ratings. In Figure 4, private rooms get rated higher than entire homes, with a mean of 4.84 and 4.70, respectively. This is counterintuitive, as renting an entire home generally costs more than a single private room. This could be because the expectations of guests who rent entire homes may be higher than those of guests who just rent a single room, resulting in a lower rating.

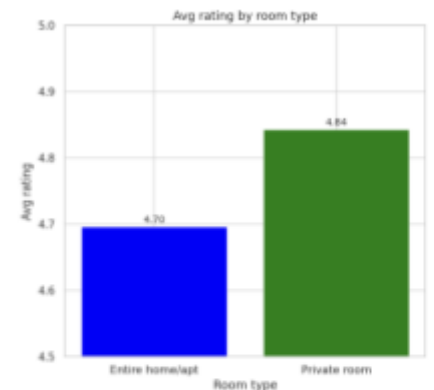


Figure 4. Bar Chart showing the relationship between average room rating by room type (entire home, private room)

Next, we analyzed the relationship between room size and ratings.

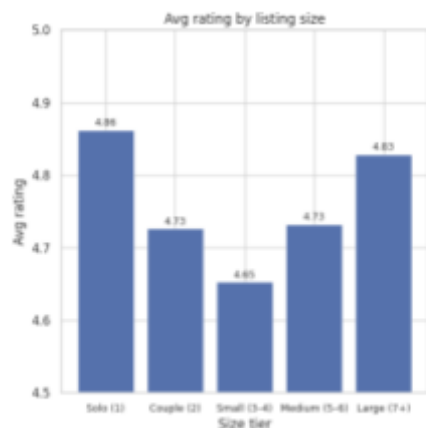


Figure 5. Bar chart showing the relationship between average rating and listing size

As seen in Figure 5, both solo listings and large group accommodations (7+ people) have higher ratings of around 4.83-4.86, while the smaller to medium-sized groups have lower ratings of around 4.65-4.73. Since larger Airbnbs that accommodate bigger parties are generally priced higher, this U-shaped pattern suggests no consistent positive correlation between accommodation size and high ratings. This could be because it is hard to match the expectations of a mid-sized party rather than a cozy solo trip or a large getaway. Finally, we explored the relationship between the number of amenities and ratings. In Figure 6, listings with many

amenities tend to have higher ratings overall, with a mean of 4.87. This suggests that one possible price proxy, amenity count, is positively associated with ratings. However, because other proxies such as room

type and accommodation size show mixed patterns, the evidence does not support a clear conclusion that more expensive listings consistently receive higher ratings.

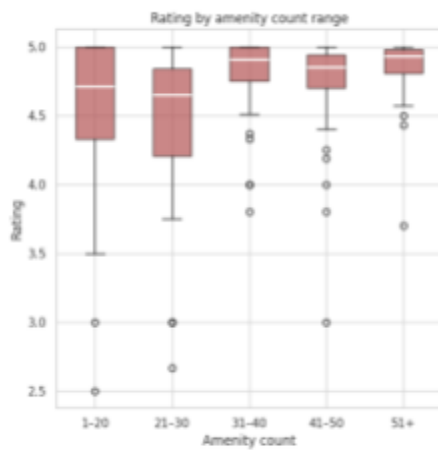


Figure 6. Box plot showing average ratings by amenity count

Overall, the analysis reinforces this finding, suggesting that more expensive listings do not consistently receive higher ratings. This could be because Airbnb ratings are shaped less by the quality or cost of a listing and more by the gap between guest expectations and their actual experience. Those who are paying for a luxury experience have higher standards and can be easily disappointed, as reflected in their reviews. In conclusion, more than price, expectations seem to be the main driver in customer ratings.

RQ 3. Which factors can a host actually change (cleanliness, response time) vs. which are fixed (location, neighborhood)?

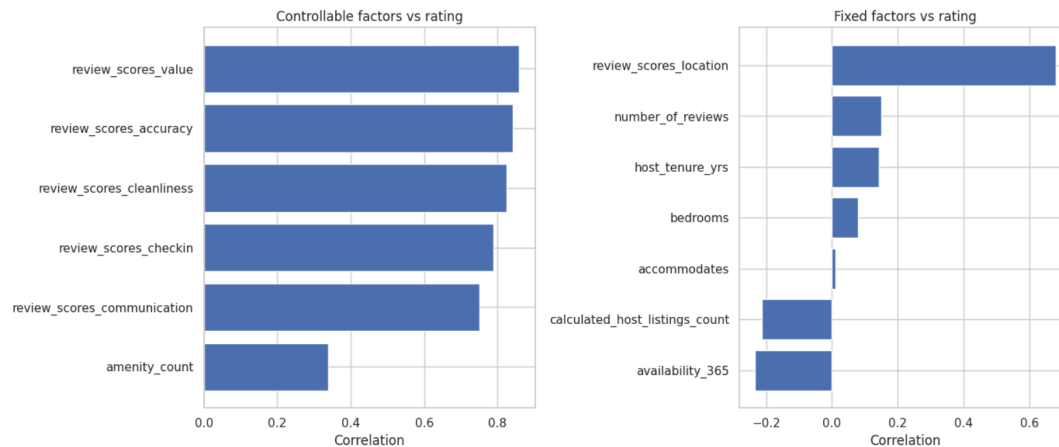


Figure 7. Pearson correlations with overall rating for actionable factors (left) and fixed or structural factors (right). Raw column names are shown as they appear in the dataset.

To address this question, features were divided into two categories: those a host can actively control and those that are structurally fixed once a listing is established. Correlation with overall rating was used as the measure of each factor's association with guest satisfaction.

Among the actionable factors, the value sub-rating ($r = 0.86$) and accuracy sub-rating ($r = 0.84$) showed the strongest associations with overall rating, meaning guests who felt a listing did not deliver on its description or its implied price point were the most likely to leave a lower overall score; both of which

are directly within a host's control through honest listing descriptions and competitive pricing. Cleanliness ($r = 0.82$) and communication ($r = 0.75$) were also strongly associated with higher ratings, reinforcing that day-to-day hosting behaviours have a meaningful and measurable impact on guest satisfaction. Check-in experience ($r = 0.79$) similarly showed a strong positive relationship, suggesting that the arrival process is a critical moment in shaping the guest's overall impression. Amenity count, while requiring upfront investment, also showed a positive association ($r = 0.34$), indicating that tangible improvements to a listing's offering translate into better ratings over time.

On the fixed side, location ($r = 0.68$) was the strongest positive correlate but is entirely outside a host's control once a property is established. Availability ($r = -0.24$) and host listing count ($r = -0.21$) both showed negative correlations, suggesting that listings with more calendar availability tend to rate lower, likely because heavily available listings reflect lower overall demand. Number of reviews ($r = 0.15$) and host tenure ($r = 0.14$) showed weak positive associations that improve only gradually over time and cannot be changed in the short term. Accommodates and bedrooms showed near-zero correlations, confirming that the physical size of a property has little bearing on guest satisfaction.

Overall, the analysis shows a clear asymmetry: the factors most strongly associated with higher ratings (cleanliness, communication, accuracy, and check-in experience) are all within a host's direct control. Fixed structural factors like location and property size contribute comparatively less to the ratings guests ultimately leave, suggesting that hosting behaviour and listing presentation are more actionable levers for improvement than physical property characteristics.

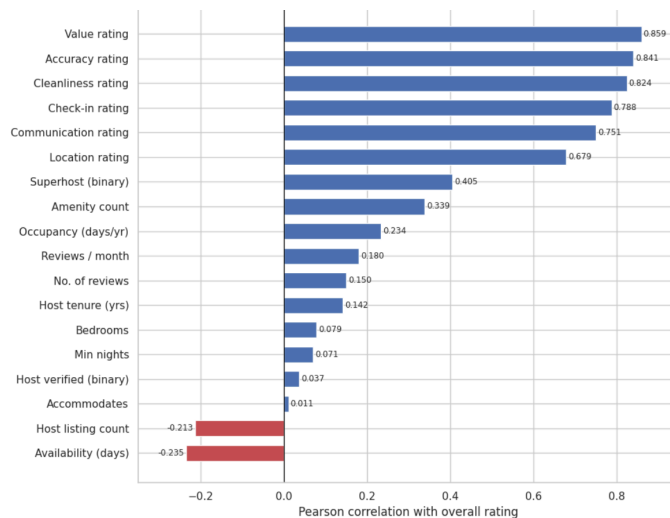


Figure 8. All features ranked by Pearson correlation with overall rating. Positive correlations are shown in blue, negative in red.

Figure 8 summarizes the complete ranking of all features by their correlation with overall rating. Sub-ratings dominate the top of the chart, with value ($r = 0.859$), accuracy ($r = 0.841$), and cleanliness ($r = 0.824$) as the three strongest predictors.

Among non-sub-rating features, superhost status ($r = 0.405$) and amenity count ($r = 0.339$) are the strongest positive predictors, while availability ($r = -0.235$) and host listing count ($r = -0.213$) are the only features with meaningful negative relationships. Features such as accommodates ($r = 0.011$) and host verified ($r = 0.037$) contribute minimally, suggesting

they have little practical value as predictors of guest satisfaction. These rankings directly informed the feature selection decisions made in the modeling phase.

Methods

Model 1: Logistic Regression

Logistic regression was selected as the first classification model due to its interpretability and its ability to produce odds ratios, which directly quantify how each feature influences the probability of a listing being high-rated. The binary target variable was defined as `high_rating = 1` if a listing's overall rating was ≥ 4.75 , and 0 otherwise. This threshold was chosen based on the rating distribution identified in the EDA, where 63.5% of listings scored in the 4.75–5.0 band, making it the most meaningful boundary for separating top-performing listings from the rest.

The feature set consisted of 11 host and listing characteristics: amenity count, superhost status, host identity verification, host tenure, number of reviews, reviews per month, availability, host listing count, accommodates, bedrooms, and minimum nights. Sub-ratings were deliberately excluded to prevent data leakage, as they are collected simultaneously with the overall rating and are consequences of the guest experience rather than independent predictors. The `instant_book` feature was excluded because it was entirely missing in the rated listings and therefore provided no usable predictive signal.

Data was split into training (80%) and test (20%) sets using stratified sampling to preserve class balance. Missing values were handled using median imputation via `SimpleImputer`, fitted on the training set and applied to the test set to prevent leakage. The model was trained with `max_iter=1000` to ensure convergence. Coefficients were exponentiated to produce odds ratios, which were used to interpret the direction and magnitude of each feature's effect on the probability of being high-rated.

Model 2: Random Forest

A random forest classifier was selected as the second model to capture non-linear relationships and feature interactions that logistic regression cannot model. A random forest is an ensemble method that aggregates predictions across many decision trees, reducing variance and improving generalization. The same feature set, train/test split, and imputation pipeline used for logistic regression were applied to ensure a fair comparison.

The model was trained with 100 estimators and a fixed random state of 42 for reproducibility. Feature importances were extracted using mean decrease in impurity, which measures how much each feature contributes to reducing uncertainty across all trees. This provides a complementary perspective to the odds ratios from logistic regression: odds ratios show directional effects, while feature importances show overall predictive contribution regardless of direction.

Model Comparison (Performance Tradeoffs):

Both models were evaluated on the held-out test set using four metrics: accuracy, precision, recall, and ROC-AUC. Accuracy measures overall correct classifications. Precision measures what proportion of predicted high-rated listings were actually high-rated, reflecting how trustworthy positive predictions are. Recall measures what proportion of actual high-rated listings were correctly identified, reflecting the model's ability to avoid missing true positives. ROC-AUC summarizes performance across all classification thresholds, providing a threshold-independent measure of discriminative ability. Predicted probability distributions were plotted for both models to assess separation between the two classes. A well-performing model should show high-rated listings concentrating toward probability 1 and non-high-rated listings concentrating toward 0. A calibration plot (reliability diagram) was also produced to assess whether each model's predicted probabilities accurately reflect true positive rates across probability bins, with a perfectly calibrated model following the diagonal.

Results & Analysis

Model Performance

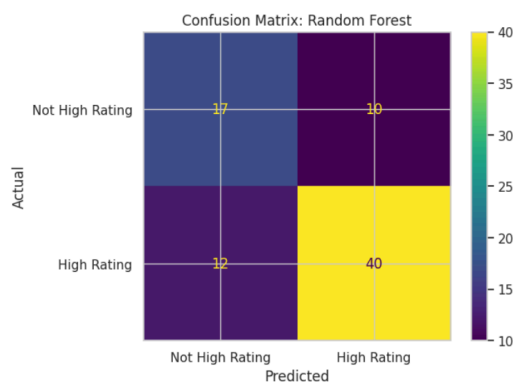


Figure 9. Confusion matrix for the Random Forest classifier. Figure 10. Confusion matrix for the Logistic Regression classifier.

Both models were evaluated on the held-out test set. The Logistic Regression classifier achieved an accuracy of 0.734, precision of 0.830, recall of 0.750, and a ROC-AUC of 0.806. The Random Forest classifier achieved an accuracy of 0.722, precision of 0.800, recall of 0.769, and a ROC-AUC of 0.802. Both confusion matrices show that the majority of misclassifications occur as false negatives; listings that were actually high-rated but predicted as not high-rated, rather than false positives. This indicates that both models are conservative in their positive predictions.

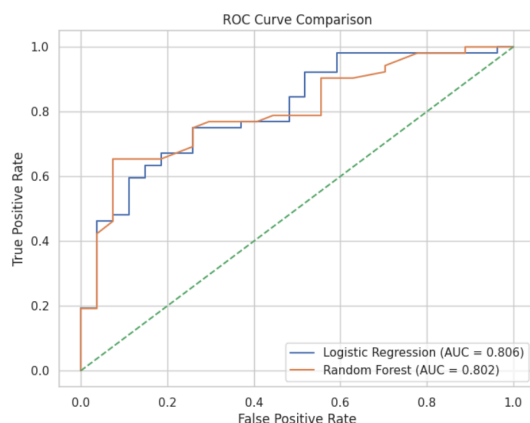


Figure 11. ROC curves for Logistic Regression and Random Forest on the held-out test set. The diagonal dashed line represents a random classifier baseline.

The ROC curves in Figure 11 confirm that both models meaningfully outperform a random baseline across all classification thresholds, with both models performing

nearly identically, and Logistic Regression marginally outperforming Random Forest on AUC (0.806 vs 0.802).

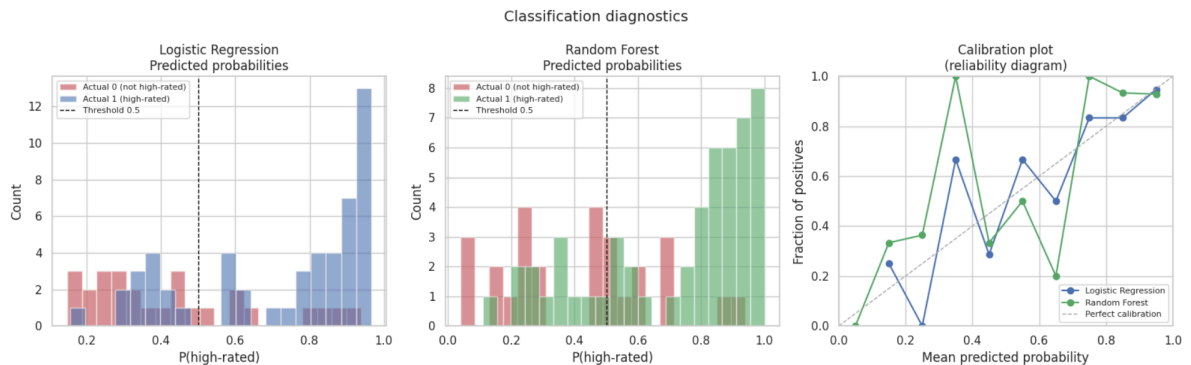


Figure 12. Classification diagnostics for both models. Left and center: predicted probability distributions separated by actual class. Right: calibration plot showing alignment between predicted probabilities and observed positive rates

The predicted probability distributions in Figure 12 show that both models produce reasonable class separation, with high-rated listings concentrating toward higher predicted probabilities and non-high-rated listings concentrating toward lower values. The Random Forest shows slightly stronger clustering of high-rated listings near higher probabilities, but its AUC is nearly identical to Logistic Regression's. Logistic Regression marginally outperformed Random Forest on ROC-AUC, while Random Forest had slightly higher recall. The calibration plot suggests approximate calibration, although variability across bins indicates some instability due to the relatively small sample size. This suggests that Logistic Regression provides slightly more reliable overall classification performance, while Random Forest is better at identifying a greater proportion of high-rated listings, reflecting a tradeoff between precision and recall.

Key Drivers of High Ratings (Feature Importance):

The Random Forest feature importance chart (Figure 13) indicates that superhost status is the most influential predictor of high ratings, followed by availability, amenity count, and reviews per month. From a guest's perspective, these are exactly the features associated with a reliable and comfortable stay. Superhosts are held to strict platform standards, while high amenity counts and frequent reviews signal a well-maintained and actively managed listing. The number of reviews also helps calibrate guest expectations, as listings with larger review samples provide more reliable signals of consistent quality.

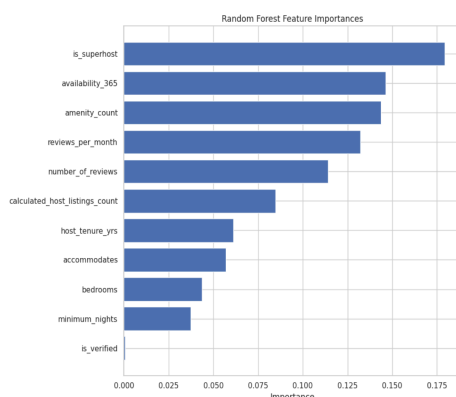


Figure 13. Random Forest feature importances ranked by mean decrease in Gini impurity. Superhost status is the strongest predictor of high ratings, followed by availability, amenity count, and reviews per month.

Figure 14. Logistic Regression odds ratios for each feature. Values above 1.0 indicate a positive association with the probability of being high-rated; values below 1.0 indicate a negative association.

Feature	Coefficient	Odds Ratio
1 is_superhost	2.468041	11.799314
9 bedrooms	0.060290	1.062145
10 minimum_nights	0.034213	1.034805
0 amenity_count	0.026550	1.026905
5 reviews_per_month	0.010228	1.010280
4 number_of_reviews	0.001516	1.001517
6 availability_365	-0.001909	0.998093
7 calculated_host_listings_count	-0.044497	0.956479
8 accommodates	-0.058873	0.942826
3 host_tenure_yrs	-0.132246	0.876125
2 is_verified	-0.286837	0.750634

The Logistic Regression odds ratios in Figure 14 complement this picture by showing the directional effect of each feature. Features with odds ratios above 1.0 increase the probability of being high-rated, while those below 1.0 decrease it. This provides an interpretable, actionable view of which host behaviours and listing

characteristics most reliably distinguish top-performing listings from the rest.

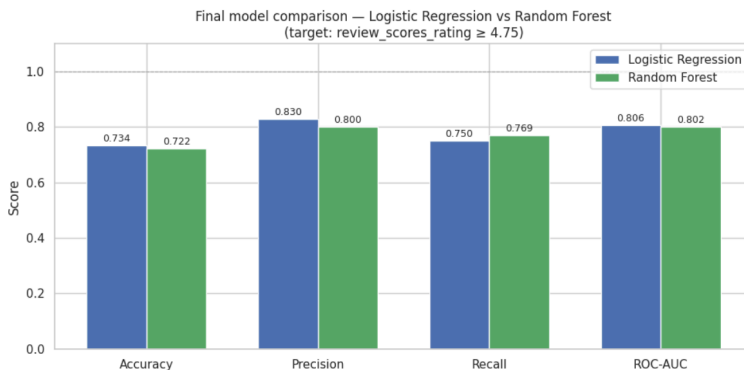


Figure 15. Final model comparison across accuracy, precision, recall, and ROC-AUC. Higher scores indicate better performance on the held-out test set.

Summary of Findings:

After extensive analysis of the dataset from Inside Airbnb, we can conclude that high Airbnb ratings are driven by a combination of strong value, cleanliness, a wide range of amenities, and superhost status. In this context, the value sub-rating reflects guests' perceived value, while room type, accommodation size, and amenity count were used as rough proxies for price because actual pricing data was unavailable. These qualities were consistently observed across multiple forms of analysis and visualizations, reinforcing their importance in shaping guest satisfaction. These findings offer a strong foundation for understanding how hosts can enhance performance and stay competitive on the platform.

Discussion

Starting with Research Question 1, having superhost status on Airbnb serves as the strongest categorical predictor of ratings, with a difference of 4.886 average stars for superhosts and 4.577 stars for everyone else. In terms of numerical correlation, amenity count had the highest correlation with higher amenity listings coming to an average of 4.87 stars versus lower amenity listings coming to an average of 4.55 stars. These findings indicate that both host-related factors and tangible listing features play a

significant role in shaping guest satisfaction. Importantly, even small differences in ratings are meaningful because most Airbnb listings receive very high scores, so even slight drops can significantly affect how a listing is perceived (Zervas et al., 2015).

Turning to Research Question 2, the analysis examined whether higher value properties led to more critical reviews. Findings showed that factors such as room type and accommodation size produce a non-linear relationship with average ratings. This suggests that ratings are influenced less by objective features of the property and more by how well the experience aligns with guest expectations. This aligns with prior research showing that ratings and reviews strongly influence customer booking decisions on Airbnb, where even small differences in quality can affect guest choice (Zervas et al., 2017).

For Research Question 3, the analysis showed that higher ratings were associated with cleanliness, communication, listing accuracy, and check-in experience. These are all factors that are controlled by the hosts themselves, which complements the conclusion from Research Question 2 that overall guest expectations play a big role in the rating of a property.

Based on these findings, there are certain insights noted based on analyzing listing data and identifying common patterns. Our first recommendation would be to encourage users to pursue superhost status, as there is a clear progression with the average ratings of a superhost being 0.31 stars higher than a normal host. There is also a clear relationship with properties with more amenities receiving higher ratings than lower amenity properties. Investing in features that enhance the guest experience is critical, as even simple yet valued amenities such as laundry and WiFi can significantly improve a host's ratings.

In addition to these factors, it can also be concluded that guest expectations play a central role in determining ratings, so ensuring that listings are accurately described and transparent is essential. The higher the expectations, the more difficult it becomes for hosts to exceed them. This reinforces the idea that, on Airbnb, ratings reflect not only service quality but also the alignment between expectations and actual experience.

Finally, the most significant limitation of this analysis was the lack of pricing data in the dataset. There are many qualitative variables that affect ratings such as host status, amenity count and cleanliness; however, it is also important to consider quantitative factors. Price is a key driver of guest expectations, and without it, the relationship between perceived value and satisfaction cannot be fully evaluated. If a smaller amenity property comes at a higher price, guest expectations may be higher, making the rating more difficult to maintain and harder for hosts to consistently meet expectations. Without determining the true extent of the effect that price has on these ratings, our analysis cannot be considered fully complete.

Future analysis should incorporate pricing data to better understand how cost influences guest expectations and overall rating behavior. Including pricing would allow for a more direct evaluation of

perceived value and help explain how guests assess whether a listing meets expectations, which is likely a key driver of satisfaction and ratings.

Overall, these insights provide a clear framework for how hosts can strategically improve their listings and remain competitive in an increasingly data-driven marketplace.

References

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